

ProjWT — WHOOP 4.0 Reverse-Engineering Build Log

Project: ProjWT (personal portfolio project — private) **Owner:** Hamza Khan **Status:** Phase 1–2 (BLE access + protocol) — in progress **Last updated:** 2026-10-01

PRIVATE / PORTFOLIO. This document and the whole ProjWT repo are a personal portfolio piece and are not to be shared publicly.

This is the **living record** of the project — every step, command, result, and decision. It is the source of truth; the PDF/Word versions are generated from this file (see **Regenerating the PDF/Word** at the end). Keep it updated as we go.

1. What we're building (and the safety line)

A personal system that talks directly to a WHOOP 4.0 band over Bluetooth Low Energy (BLE), reads its sensors, and eventually drives:

- **Biometric dashboard** — live heart rate, HRV, SpO₂, skin temp, motion.
- **Gesture control** — recognise wrist gestures from the band's accelerometer.
- **Vehicle control (Tesla Fleet API)** — *supervised only.*
- **Home / IoT automation.**

Target stack (eventual): C++ / Unreal Engine 5. **Phase-1 tooling (now):** Python — fastest way to reverse-engineer and validate the protocol before committing it to C++.

Safety constraint (non-negotiable)

Vehicle control is **active-hold, line-of-sight only**:

- You hold a gesture **while watching the car**; the car moves only while the gesture is actively held.
- Releasing the gesture (a dead-man's-switch) slows or stops the car.
- **No** "no-line-of-sight summon", **no** GPS spoofing, **no** safety-bypass.

That idea was considered and **dropped**.

2. The hardware target

- **Device:** WHOOP 4.0 band (owned by the project author).
- **Advertised name:** WHOOP_4C1656042
- **BLE address (Windows, this session):** D1:86:73:D4:62:86
- ■■ On Windows this address can be a rotating/resolvable random address, so **re-scan each session** rather than hard-coding it (see §8).

- **Battery at bring-up:** 100% (band healthy, charged via the pack; tap → LED).

3. Environment setup (Windows)

The band talks to the PC directly over BLE — **no WHOOP app, no pairing needed**. We drive it from Python.

- 1 **Install Python** (3.10+). During install tick **"Add Python to PATH."**

- 2 **Install the BLE library:**

```
python -m pip install bleak
```

- 1 **Open a terminal in the code folder.** In File Explorer, open the projwt\src folder, click the address bar, type powershell, press Enter.

- 1 Run scripts with python first. PowerShell will ***not*** run ble_scanner_PWT.py on its own (that just opens Notepad). Always:

```
python ble_scanner_PWT.py --name whoop
```

Common beginner errors we already hit and fixed - **"The term 'ble_scanner_PWT.py' is not recognized"** → you forgot the python prefix. Start the command with python. - **"Pressing Enter opens Notepad"** → same cause; Windows is "opening" the file instead of running it. Use python <file>. - **"ModuleNotFoundError: No module named 'bleak'"** → run python -m pip install bleak.

4. The tools we built (in src/, all _PWT)

Script	Purpose
ble_scanner_PWT.py	Discover BLE devices, print name / address / RSSI. Find the band.
gatt_explorer_PWT.py	Enumerate the band's GATT services / characteristics / descriptors.
ble_logger_PWT.py	Subscribe to notify channels and log raw packets to logs/ (JSONL). --label, --seconds.
whoop_protocol_PWT.py	The verified protocol core — framing, CRCs, command IDs, UUIDs.
ble_command_PWT.py	Send one framed command and listen for the reply.
ble_stream_PWT.py	HELLO → START → watch data channel for a continuous stream.
ble_sweep_PWT.py	Sweep a range of command IDs to find which one wakes the data channel.

Script	Purpose
<code>ble_pair_PWT.py</code>	Pair/bond with the band so it will accept our commands (encrypted link).
<code>ble_stop_PWT.py</code>	Send STOP (0x04) to switch the realtime stream + green HR LED back off.
<code>ble_startseq_PWT.py</code>	Try several start sequences in one connection to find the stream trigger.
<code>ble_accel_PWT.py</code>	Find/stream the accelerometer (payload sweep, opcode sweep, live decode).
<code>ble_events_PWT.py</code>	Capture & decode live EVT (04) events, tagged by gesture label.
<code>gesture_PWT.py</code>	Live gesture recognition — tunable FLICK trigger from 0x0e impulses (+ strength filter).
<code>safety_state_machine_PWT.py</code>	L3 fail-closed safety core — pure logic, 7 unit tests, no hardware.
<code>decode_realtime_PWT.py</code>	Decode a saved capture into heart rate + RR intervals (and CSV).
<code>decode_debuglog_PWT.py</code>	Mine the band's own firmware debug-log strings (type 0x32) from a capture.
<code>ble_rawprobe_PWT.py</code>	Try the real raw/IMU enable commands (0x6a/0x6b/0x3f) and watch for live raw (0x2b/0x33).

Everything is committed to the private repo Hamza-Khan-algonquin/projwt, branch `claude/peaceful-dirac-ko6o1i`.

5. What we did, step by step (chronological)

- Charged and woke the band.** Tapping it showed the LED; battery reached 100%. Windows Bluetooth couldn't see it at first (it wasn't advertising / the WHOOP app's own pairing kept crashing). We bypassed all of that and scanned directly from Python.
- Scanned with `ble_scanner_PWT.py`. Found WHOOP 4C1656042 at D1:86:73:D4:62:86, RSSI $\approx$ -61 dBm (strong, close).
- Mapped the GATT table with `gatt_explorer_PWT.py`. Got the custom WHOOP service and its characteristics (see §6).
- Logged raw notifications with `ble_logger_PWT.py`. **Got 0** packets at rest — **expected**: the band stays silent until told to stream. Also fixed the ugly Ctrl+C traceback and added - - seconds auto-stop.

- 1 **Reverse-engineered the packet framing.** Community write-ups disagreed with each other (they were paraphrased summaries), so we **brute-forced the CRC parameters against a real captured packet inside a sandbox** until the framing reproduced byte-for-byte. This is now the verified core (§6).
- 1 Sent our first real command with `ble_command_PWT.py`. **The band** replied with a valid, CRC-correct packet — **proof our framing is accepted by real hardware (two-way comms confirmed)**. **The reply came on the** event channel (61080003), 32 bytes, a protobuf-style status/report — **not** a continuous sensor stream.
- 1 **Diagnosed why there was no stream.** Reference implementations perform a HELLO handshake (cmd 0x05) first, **then START; the continuous data arrives as ~96-byte packets on the** data channel (61080004). We had skipped the hello. Built `ble_stream_PWT.py` to do the full sequence.
- 1 Ran the full HELLO → START sequence (`ble_stream_PWT.py`). The band came alive — **15 packets in 30s** — but **all on the EVENT channel (61080003)**, a repeating ~2s **status beacon**, **not** the 96-byte sensor stream on the DATA channel (61080004, which stayed silent). Two outcomes:
 - **Decoded the beacon's identity fields** (see §6.5): codename `boylston`, firmware **17.2.2.0**, hardware `harvard_r10`.
 - **Found & fixed a display bug:** the tools labelled channels by the **last** 8 UUID chars, but every WHOOP characteristic ends in `...b0f8dcc6`, so all channels printed the same label. They differ in the **first** block (6108000x); labelling now uses a clear name map (`EVENT(03)`, `DATA(04)`...).
- 1 Concluded cmd 0x03 is not the real "start streaming" opcode (the command IDs were always community guesses). Built `ble_sweep_PWT.py` to brute-force the correct opcode — the same empirical approach that cracked the CRC framing.
- 1 Swept command IDs (`ble_sweep_PWT.py`) — and learned two big things:
 - **The band ignores unauthenticated commands.** Write-**with**-response returned GATT Protocol Error: Insufficient Authentication (ATT error 0x05 — the command characteristic needs an **encrypted / paired link**). Write-**without**-response (our default) doesn't error, but the band **silently ignores it**. So **none** of our commands had ever actually reached the band* — every "reply" so far was the subscribe-triggered status beacon or physical motion.
 - **DATA(04) is currently a motion/tap event feed.** The DATA-channel packets lined up with **tapping / shaking / putting the charger on**, not with any command. They use our verified 0xAA framing, type 0x30, **with an incrementing sequence and a monotonic** timestamp — i.e. timestamped accelerometer/activity events (useful later for gesture detection).
- 1 Built `ble_pair_PWT.py` and added `--pair` to the sweep/stream tools.
Pairing establishes the encrypted link so the band will honour commands.
- 1 **PAIRED — and the band now answers our commands.** After pairing, a sweep

with write-with-response showed no more Insufficient Authentication, **and commands** 0x01–0x04 each returned a genuine 32-byte type-0x30 reply on DATA(04) (confirmed: band untouched, timestamps incrementing 0x01e28a92 → ...95 → ...98 → ...9c). cmd 0x0b was a physical tap.

Two-way authenticated command/response is working.

1 **Found the real cause — we were watching the WRONG channel.** Comparing the reference (christianmeurer/whoop-reader) char constants against our own scan showed an **off-by-one**: the reference assumes the write char is ...0001, but on our band ...0001 is the **service** and the write char is ...0002, so every characteristic is shifted +1. Aligning by role (write + 4 notify, in handle order) means the real-time sensor stream is on 61080005 — **the channel we had dismissed as "diagnostics". What we called "DATA(04)" is actually the *event* channel (hence one-off command acks + taps). See §6.1 for the corrected map. Also: the reference's start sequence is identical to ours (subscribe → HELLO 0x05 → START 0x03, no payload, response=True), so the opcode was never wrong — the channel was. And every test so far was off-wrist (WHOOOP gates the PPG/HR stream on skin contact).**

1 Fixed the channel map (DATA = 61080005), solved the Windows idle-drop with a harmless Battery-Level-read keepalive, and built ble_startseq_PWT.py to try multiple start sequences in one stable connection.

1 **■ REAL-TIME STREAM UNLOCKED.** The start command is 0x03 with payload 0x01 (bare 0x03 does nothing on fw 17.2.2.0). Data then streams on DATA(05) as ~1 Hz type-0x28 packets **carrying** heart rate + RR intervals. Decoded and verified: one packet reported HR 76 with a single RR of 789 ms, and $60000/789 = 76.0$ exactly. Captured HR climbing 67→78 bpm with RR ~740–820 ms. Built decode_realtime_PWT.py to decode captures to HR/RR/CSV.

1 HR validated against the owner's real resting BPM — decode confirmed.

2 **ACCELEROMETER FORMAT FOUND.** Probing opcodes 0x10–0x1f (ble_accel_PWT.py --cmd-sweep) unlocked a high-rate burst of ****96-byte**** type 0x2f ****packets**** on DATA(05). Decoded: ****accelerometer X/Y/Z as three float32 in g units**** at payload offset 36 (a second copy at 52), e.g. a rest vector (-0.125, 0.366, 0.934) → magnitude ****1.01 g**** (gravity). The same window also emitted firmware ****debug-log text**** (type 0x32) and the dump was labelled "Historical Dump" — so 0x16 downloads STORED data. Next: confirm the same 0x2f float format arrives LIVE while moving the wrist.

1 **Op notes:** Windows BLE needs a Bluetooth off/on toggle to recover from the "operation was canceled" wedge, and the band must be tapped awake right before running. Fixed a tool crash (formatting unknown packet types) and added live accel/HR decoding + --extra-cmd.

1 Accel from 0x16 is HISTORICAL, not live. Those 0x2f packets carry a real calendar timestamp (decoded: 2024-09-18), whereas live packets use device-uptime ticks. Re-running 0x16 moving the wrist produced no new accel — 0x16 downloads stored flash records. **Conclusion:** on fw 17.2.2.0, raw accelerometer is recorded to flash and retrieved as history; it does not appear to stream live over BLE via opcodes 0x01–0x1f. Confirmed LIVE signals: HR+RR (type 0x28, 1 Hz) and discrete events on EVT(04) (type 0x30).

1 Swept 0x21–0x2f (skipping 0x20 firmware) — no live accel. Every opcode

returned only the baseline HR stream (type 0x28); no 0x2f, no new subtype, no rate jump.

CONCLUSION (firmware 17.2.2.0): raw accelerometer is NOT exposed as a live BLE stream anywhere in opcodes 0x01–0x2f. It exists only as historical flash records (via 0x16). Live BLE signals available: HR+RR (0x28, 1 Hz) and discrete events on EVT (04) (0x30).

1 **Design implication:** a smooth continuous "active-hold" wrist gesture isn't

possible from the band alone over BLE (no live orientation stream). Discrete gestures (tap / double-tap / motion triggers) via EVT (04) are, plus live HR.

1 **Live events characterized (labeled captures).** Ran `ble_events_PWT.py` for

still / tap / doubletap / flick / raise / armswing. Result:

- 0x0e = **impulse event** (sharp motion): still=0, tap=5, doubletap=6, flick=9, raise=0, armswing=0. It fires on taps/flicks and **ignores slow or large motion** — slow raise and vigorous armswing produced *zero* 0x0e (only HR rose). Green LED = tap registered; taps need to be sharp, flicks are the most reliable primitive.
- 0x21 = once-per-session status/startup event (not a gesture).
- 0x03 / 0x3f = occasional richer "activity report" events (multi-field).
- Confirms: the only live motion signal is **impulse detection**, not continuous orientation.

1 **Built the live gesture recogniser** (`gesture_PWT.py`): clusters 0x0e

impulses in time → TAP (1) / DOUBLE_TAP (2) / FLICK (≥ 3), with on-screen output and a stable session. This is the L2 discrete-gesture engine.

1 **Impulse strength field confirmed + firmware floor is real.** The 0x0e body's

first int16 is an **impact-strength** value: a real flick's initial impulse reads ~7000–31000, while the ringing/settle that follows reads ~1400–4800 (clear gap). So each hard flick can be reduced to one clean FLICK (strength filter + burst grouping + cooldown; defaults: `--strength 6000 --cooldown 0.6`).

But the "must flick hard" requirement is a FIRMWARE floor — the band only emits the event above its own threshold; no host setting lowers it (confirmed; no sensitivity command exists). A host strength filter only improves *accuracy*, not *sensitivity*.

1 **Design note:** a deliberate firm flick is acceptable — even desirable — as a

discrete **arm / confirm** trigger (you don't want an accidental light motion firing a vehicle/IoT command). Continuous "active-hold" still needs the 2nd sensor per the architecture; discrete confirm = hard flick from the band is fine.

1 **Built the L3 safety state machine** (`safety_state_machine_PWT.py`) — pure,

hardware-free, fail-closed logic (IDLE/ARMED/HOLDING/STOPPING) consuming gesture (FLICK=arm/disarm), hold (continuous source), heartbeat, tick, and emitting guarded ARM/DISARM/MOVE/STOP commands with a TTL on MOVE. **7 unit tests pass** covering: happy path, continuous-hold MOVE cadence, hold-without-arm (nothing), disarm-stops, arm-timeout, heartbeat-loss stop, and STOPPING can't jump back to HOLDING. Runs with `python safety_state_machine_PWT.py`.

1 **Firmware-config lead (sensitivity):** the band's own debug-log text (type

0x32, captured during the 0x16 historical dump) includes Sensors: Realtime HR disabled and Sensors: Realtime raw disabled — i.e. a **raw sensor mode that is off by default**. Enabling it over BLE (a config command, NOT firmware patching — the image is signed/brick-risk) could give live raw accel and make the tap threshold irrelevant. Built `decode_debuglog_PWT.py` to mine those strings for the command vocabulary. (Binary firmware patching is out of scope: signed, needs SWD/JTAG + key, bricking risk.)

1 **Found the raw/IMU enable commands via community RE** (tanarchytan/whoop-rs, MIT). The enables live ABOVE our swept range: `SET_IMU_DATA_STREAM 0x6a [01, 01]` (live 100 Hz IMU → type 0x33) and `SEND_OPTICAL_DATA 0x6b [01, 01]` (raw optical + v21 IMU → some live as type 0x2b). Both session-scoped/safe. Built `ble_rawprobe_PWT.py` to enable each and watch for live 0x2b/0x33. See §6.8c.

1 ← **You are here.** Run `ble_rawprobe_PWT.py` worn + moving → confirm live IMU on Gen4. If it works: real live accel ⇒ proper gesture/continuous-hold from the band itself. If not (R22/Gen5-gated): fall back to FLICK + phone-IMU per architecture.

6. The verified protocol (the important part)

6.1 GATT map (from our own scan — authoritative, roles corrected)

Our band has the **service** at ...0001, so the characteristics are shifted +1 versus community write-ups that assume the *write* char is ...0001. Mapping by role (write + 4 notify, in handle order):

UUID (first block)	Role	Notes
61080001	Service	custom GATT service (not a characteristic)
61080002	Command (write)	where we send framed commands
61080003	Response / status	command responses + the ~2s status beacon
61080004	Events	async events — command acks, tap/motion (type 0x30)
61080005	DATA — real-time stream	the 96-byte sensor packets live here
61080007	Diagnostics	
00002a37	Standard Heart Rate	standard BLE HRM characteristic

We originally mis-labelled 61080004 as "data" and 61080005 as "diagnostics" (by matching the reference's numbers instead of its roles). The +1 shift means the real sensor stream is on 61080005. Corrected in `whoop_protocol_PWT.py`.

6.2 Packet framing — empirically verified

0xAA | length(2, little-endian) | CRC8(length) | PAYLOAD | CRC32(PAYLOAD, LE)

```

length = len(PAYLOAD) + 4          (payload plus the trailing CRC32)
CRC8    = poly 0x07, init 0x00, no reflection, over the 2 length bytes
CRC32   = standard zlib.crc32 over PAYLOAD, stored little-endian
PAYLOAD = [type][seq][cmd][data...]
type    = 0x23 for commands

```

Proof: `build_packet()` reproduces the known-good reference packet exactly:

```

aa100057230423aa8ed469a96d0000005130fef3  ← reference
aa100057230423aa8ed469a96d0000005130fef3  ← our build_packet()  ✓ MATCH

```

Run the self-test any time: `python whoop_protocol_PWT.py`.

■■ The MIT reference repo uses a **different, simpler** framing (no CRC8). We do **not** use it — **our** framing is the one the real band actually replied to.

6.3 Command IDs (community-sourced; being verified live)

Cmd	Meaning
0x01	Get battery
0x02	Get device info (fw / serial / hw)
0x03 + payload 0x01	Start real-time streaming (bare 0x03 does nothing on fw 17.2.2.0)
0x04	Stop real-time streaming
0x05	HELLO / handshake

6.4 Real-time data packet — VERIFIED (type 0x28, ~1 Hz, on 61080005)

After `START = 0x03 01`, the band streams one ~28-byte frame per second on the DATA channel. Our 0xAA framing applies; the **payload** (after unframing) is:

Payload bytes	Field	Notes
[0]	packet type	0x28
[1]	subtype	0x02 = realtime metrics
[2:6]	timestamp	uint32 LE, device uptime seconds
[6:8]	aux	uint16 LE — not yet decoded (PPG/activity?)
[8]	heart rate	bpm (uint8)
[9]	N	number of RR intervals that follow
[10 : 10+2N]	RR intervals	N × uint16 LE, milliseconds (beat-to-beat → HRV)

Consistency check: a packet with HR=76 carried a single RR of 789 ms, and $60000 \div 789 = 76.0$. A capture showed HR climbing 67→78 bpm with RR ~740–820 ms. Decoder: `python decode_realtime_PWT.py <logfile>`.

This replaces the earlier **tentative** 96-byte layout from the reference — our firmware streams these compact 28-byte HR/RR packets instead.

6.8 Accelerometer packet — type 0x2f (96 bytes, on 61080005)

Unlocked by probing opcodes 0x10–0x1f. The accelerometer is reported as **float32 (little-endian)**, in g units:

Payload bytes	Field
[0]	packet type (0x2f)
[1]	subtype (0x0c)
[36:40] [40:44] [44:48]	accel X / Y / Z — float32, g units
[52:64]	second X/Y/Z copy (filtered vs raw)

Verified: a rest packet decoded to (-0.125, 0.366, 0.934), magnitude **1.01 g** (gravity). Parser: `whoop_protocol_PWT.parse_accel()`; live view: `python ble_accel_PWT.py <ADDR> --start-data 01 --extra-cmd 0x16 --label tilt`.

type 0x32 packets seen alongside are firmware **debug-log text** (ASCII), not sensor data. **Important:** the 0x16 dump is HISTORICAL — these 0x2f packets carry a real calendar timestamp (2024-09-18), so the accel values are stored flash records, not a live feed. Live raw accel has not been found over BLE on this firmware (opcodes 0x01–0x1f); see §9 for the gesture-control fork.

6.8b Firmware internals (from the band's own debug-log strings)

Mined from type 0x32 debug text during a 0x16 historical dump (`decode_debuglog_PWT.py`):

- **Two realtime modes, independently toggled:** `Sensors: Realtime HR` and `Sensors: Realtime raw` (both logged as disabled when idle). We drive **HR** via 0x03 01; `Realtime raw` is the target for live accel/PPG but its enable command is unknown (the 0x03 payload only toggles HR).
- **Config system exists:** `CONFIG_VALUE_OFF` (settable config enum).
- **Internal stream/packet IDs:** `R7 realtime stream`, ``R10+R11 data packet transmission default``.
- **BLE:** `Command Send Historical Data = our 0x16; dump prints `History burst success ... Trim: 0x... and PullStats: Data: N, Events: M, Bytes:...``.
- **HELLO** reply carries `FG SOC (tenths) (state-of-charge)` and ``Nordic Ver: 17.2.2.0``.

Open question: is `Realtime raw` exposable over BLE to a third party, or is it app-auth-gated / flash-only? Not determinable by black-box probing; the definitive answer is a BLE sniff of the official app (HCI snoop log). See §9.

6.8c Command table + raw/IMU enable (from community RE)

The MIT project github.com/tanarchytan/whoop-rs documents the full command set and confirms our framing exactly (Gen4: 0xAA, CRC8 over length, inner [type][seq][cmd], zlib-CRC32). Key

opcodes we had NOT reached (our sweep stopped at 0x2f):

Opcode	Name	Payload	Effect
0x03	TOGGLE_REALTIME_HR	[01]	HR/RR stream (what we use)
0x3f (63)	SEND_R10_R11_REALTIME	[00]	richer realtime stream
0x6a (106)	SET_IMU_DATA_STREAM	[01, state]	live IMU (accel+gyro), 100 Hz 6-axis
0x6b (107)	SEND_OPTICAL_DATA	[01, state]	raw optical v20 (25 Hz) + v21 IMU; session-scoped, no flash write

Live packet types (first payload byte): 0x28 REALTIME_DATA(HR), 0x2b REALTIME_RAW_DATA, 0x30 EVENT, 0x31 METADATA, 0x32 HISTORICAL, 0x33 REALTIME_IMU_STREAM. IMU scale: accel = $\text{int16} \times 1/4096$ g, gyro = $\text{int16} \times 2000/32768$ dps. (We actually saw type 0x2b.07 during the earlier opcode sweep and didn't recognise it.) 0x6a/0x6b/0x3f are **not** in whoop-rs's forbidden/ destructive lists → safe to probe. **This is the real path to live accel** — probed by `ble_rawprobe_PWT.py`. Caveat: the deepest v20/v21 buffers may want the band worn/asleep and R22 (Gen5-only); Gen4 live support is what the probe tests.

6.9 Live events — type 0x30 on EVENT_CHAR (61080004)

Discrete events. Layout: [0]=type 0x30, [1]=seq, [2]=**report id**, [3]=0, [4:8]=uptime ts, [8:]=body. Report ids observed, from labeled captures:

Report id	Meaning	Fires on
0x0e	Impulse (tap / flick) — the gesture primitive	sharp wrist motion only
0x21	Session status / startup	once on connect
0x03, 0x3f	Activity report (multi-field body)	occasionally during motion

Key behaviour: **slow or large motion produces no event** (a slow arm-raise and a vigorous arm-swing both yielded zero 0x0e — only heart rate rose). So the band's only live motion signal is impulse detection. Gesture vocabulary is built by clustering 0x0e in time (`gesture_PWT.py`): 1 → TAP, 2 → DOUBLE_TAP, ≥ 3 → FLICK.

Captured **event** packet (32 bytes on 61080003, protobuf body) for reference:

```
aa 1c 00 ab 30 06 1d 00 2d 48 e1 01 20 1a 0c 00 32 01 00 00 02 00 42 02 0b 00 01 00 8c 60 8f ce
```

6.5 Status beacon (EVENT channel 61080003) — decoded identity

When we subscribe, the band emits a ~97-byte **status beacon every ~2 seconds** on the EVENT channel. It is a protobuf body (starts 08 02 a6 02 ...) carrying the device identity plus a changing counter near the tail. Decoded ASCII fields:

Field	Value
Codename	boylston
Firmware / version	17.2.2.0
Hardware revision	harvard_r10

This is a **heartbeat/status** feed, not the per-sample sensor stream. The per-sample sensor data (PPG / accel / HR) is expected as ~96-byte packets on the DATA channel 61080004, which is still silent — see §9 next steps.

6.6 Authentication requirement (the key blocker)

The command characteristic (61080002) requires an **encrypted/paired BLE link**. Evidence: write-with-response returns GATT Protocol Error: Insufficient Authentication (ATT error 0x05). Write-without-response doesn't raise, but the band silently discards the write. **Therefore commands only take effect after we pair/bond** (ble_pair_PWT.py, or --pair). Until then the band only emits the status beacon and motion events, regardless of what we send.

6.7 DATA(04) event packets (motion-triggered) — framing confirmed

The DATA-channel packets observed so far are our 0xAA frame with type 0x30:

```
aa | len(2 LE) | crc8(len) | 30 | seq | .. | timestamp(uint32 LE) | .. | crc32
```

Example: aa 10 00 57 30 24 0e 00 c8 87 e2 01 80 44 00 00 ... → len=16, crc8 OK, type=0x30, seq=0x24, timestamp=0x01e287c8 (monotonic). These fire on tap/shake/charger — accelerometer/activity events, not the continuous PPG stream we still need to unlock (via pairing + the right opcode).

7. Legal / attribution — do the referenced authors need to know?

Short answer: no obligation to notify anyone, and you don't have to publish that you used their work.

- The reference we consulted, christianmeurer/whoop-reader, is MIT-licensed.

The MIT license lets you use, copy, modify, and even redistribute the code — including in a closed/private project — **with one condition**: *if you redistribute their source or substantial portions of it, you must keep their copyright notice + the MIT license text with it.*

- **You are not "redistributing" it.** We only **read** it for reference and wrote

our **own** implementation (in fact with *different, independently verified* framing). So the MIT condition isn't even triggered.

- **The author is never automatically notified.** GitHub doesn't tell someone when

you read, clone, or reference their repo. There's no tracking, no callback.

- **Do you *have* to "let it be known"?** No. But crediting them is good

practice and costs nothing — so we've added a short credit note in whoop_protocol_PWT.py and here. It also makes the portfolio piece look more professional (shows you research prior art and respect licenses).

- The other names in the space (jogolden/whoomp, bwanShiTong/reverse-engineering-whoop) — same logic. jogolden/whoomp couldn't be added to this session (not found/accessible), so we used the MIT whoop-reader instead.
- Bottom line:** keep the credit note (already added), don't paste their source verbatim into your repo, and you're completely fine — legally and ethically.
-

8. Reconnecting after a break (start-of-session checklist)

The band is stateless from our side — nothing persists on the PC. Next time you sit down:

- 1 **Charge the band** if it's been idle a while; tap it to confirm the LED.
- 2 **Open PowerShell in the code folder** — File Explorer → projwt\src folder
→ type powershell in the address bar → Enter.

- 1 **Re-scan for the band** (the address may have changed):

```
python ble_scanner_PWT.py --name whoop
```

Copy the address it prints (e.g. D1:86:73:D4:62:86).

- 1 **(Optional) confirm the GATT map is unchanged:**

```
python gatt_explorer_PWT.py <ADDRESS>
```

- 1 **Pair once** (only needed the first time, or after --unpair; the bond persists across sessions):

```
python ble_pair_PWT.py <ADDRESS>
```

- 1 **Find the stream opcode** (current task — see §9), band held still:

```
python ble_sweep_PWT.py <ADDRESS> --pair --with-response
```

Once we know the right opcode, stream with:

```
python ble_stream_PWT.py <ADDRESS> --pair --with-response --seconds 30
```

- 1 Send Claude the newest file in logs\ so we can decode/continue.

If bleak is missing again (new machine): `python -m pip install bleak`. If nothing is found in the scan: make sure the band is charged and near the PC, and that no other app (the WHOOP phone app) is holding the connection.

9. Current status & next steps

Done

- Physical bring-up, BLE discovery, full GATT map.
- Packet framing reverse-engineered **and verified both ways** (we sent a command and the real band returned a valid CRC-checked reply).
- Confirmed two-way comms live; decoded the device-identity status beacon (firmware 17.2.2.0, hardware harvard_r10).

- Tooling for scan / explore / log / command / stream / sweep.

Done (new)

- **Pairing works** (Windows ConfirmOnly ceremony); authenticated link accepts commands.
- **Windows idle-drop solved** with a Battery-Level-read keepalive.
- **Real-time HR + RR streaming works** — START = 0x03 01, data on DATA(05) as ~1 Hz type-0x28 packets; decoded and verified against the owner's resting BPM.
- **Accelerometer format decoded** — type-0x2f packets, X/Y/Z float32 (g).

Finding: live raw accel is NOT available over BLE (swept 0x01–0x2f). Accel is historical-only. So gesture control uses the live EVT(04) events + HR.

Done (new): live events characterized (0x0e = tap/flick impulse) and a live gesture recogniser built (gesture_PWT.py: TAP / DOUBLE_TAP / FLICK).

Next

- 1 **Tune the recogniser on-wrist** (tap the band awake first):

```
python gesture_PWT.py <ADDRESS> --seconds 120
```

Flick a few times, tap sharply, double-tap. Adjust --gap (cluster window) and --flick (impulses for a flick) until TAP vs FLICK feel right. Send Claude a couple of logs\events_*.jsonl so we can decode the 0x0e body (impact field).

- 1 Build the **L3 safety state machine** (unit-tested, no hardware) driven by the recogniser: e.g. FLICK = arm/disarm, with fail-closed guards.
- 1 Decode the HR aux field; decode historical accel (0x2f) offline.
- 2 Port the verified protocol to C++ for the UE5 phase.

Windows gotchas (keep handy)

- "operation was canceled by the user" / repeated drops → toggle Bluetooth OFF/ON.
- Tap the band awake immediately before running any script.
- **Green HR LED stays on after a run?** START (0x03 01) powers the optical HR sensor (green LED); tools send STOP (0x04) on a clean finish, but a Ctrl+C / closed window skips it, so the band keeps streaming (battery drain, LED on even off-wrist). Fix: python ble_stop_PWT.py <ADDRESS> (or just let it idle/ disconnect and it stops on its own). Behaviour of the band, driven by our last command — not a fault.

9.9 Raw sensor data on Gen4 — what's real, and the bug that hid it

Goal of the night: get live/raw sensor data (not just HR) off the WHOOP 4.0.

Finding 1 — the 100 Hz IMU firehose is Gen5-only. Cross-checked against the MIT tanarchytan/whoop-rs decoders: the raw 100 Hz 6-axis IMU buffer (v21) and the 6-channel raw optical buffer (v20) are documented as "5.0 / MG, shipped in the R22 deep buffers."* Gen4's historical record set (records/gen4.rs) is v24, v25, v5 — **no v21/v20**. So a continuous 100 Hz accel/gyro

stream off a 4.0 is not exposed by this protocol. That's a hardware/firmware boundary, not a bug.

Finding 2 — Gen4 DOES bank real raw data; we were fetching it wrong. Gen4 records carry a lot: v24 = HR, R-R, **gravity vector** (accel-derived orientation), SpO2 (raw red/IR ADC), skin-temp, respiration; v25 = **PPG optical waveform + gravity**. These come over the *historical offload* path, and our first probe never triggered it. The correct sequence (from `whoop-rs-client.rs + offload.rs`) is:

Step	Write	Why
1	SEND_OPTICAL 0x6b [01, 01]	enable raw optical collection (no ACK is normal)
2	SEND_HISTORICAL 0x16 [00]	kick the drain — the step we were missing
3	per METADATA HistoryEnd -> HISTORICAL_RESULT 0x17 [01]+end_data	ACK each chunk so the strap advances instead of stalling
4	stop on METADATA HistoryComplete; then SEND_OPTICAL [01, 00]	finish; leave the band as we found it

Records arrive as type 0x2f HISTORICAL_DATA; the version is in the seq byte (`inner[1]`): 24/25/5. `decode_historical()` in `whoop_protocol_PWT.py` decodes HR (`inner[17]`), gravity (i16/16384 at `inner[36]` v24 / `inner[69]` v25), SpO2, skin-temp — all inner-relative, matching the Gen4 offsets pinned to real 4.0 captures upstream.

Packet-type bug fixed. We had `PKT_HISTORICAL = 0x32`; the real value is 0x2f (47). 0x32 (50) is `CONSOLE_LOGS` — which is exactly the firmware debug-string packet `decode_debug_log_PWT.py` reads, now correctly labelled.

Also tested (Phase A long-shot): HR-realtime-on -> `SET_IMU_DATA_STREAM 0x6a [01, 01]` to see whether live IMU (type 0x33) flows while realtime HR runs (the flash path always sends them in that order). Expected to stay HR-only on a 4.0; captured either way.

Tool: `ble_rawstream_PWT.py <ADDR>` — runs Phase A (live attempt) then Phase B (offload), decodes records live, logs everything to `logs/rawstream_*_PWT.jsonl`.

CONFIRMED on hardware (2026-10-04). One run drained **2429 per-second records (v12, full DSP)** spanning ~38.9 min. Decoded and validated end-to-end: HR 42–80 bpm (mean 50); gravity vectors with **99% of |g| within 0.9–1.1 g** after the float32 fix; SpO2 raw red/IR paired (~575/576); `skin_temp_raw` 903 → **36.1 °C**; respiration raw present. Pipeline: `ble_rawstream_PWT.py` (capture) → `decode_rawstream_PWT.py` (→ per-second CSV) → `plot_rawstream_PWT.py` (→ offline self-contained HTML dashboard, no external requests, data never leaves the machine). Phase A stayed HR-only (type 0x28) with no live 0x33/0x2b — live IMU firehose confirmed absent on Gen4, as predicted. Absolute timestamps are the band's own (unset) clock; per-second ordering is exact.

Takeaway for the project: continuous-hold stays on the **phone IMU** (always the plan — band motion is coarse/latent). The band's jobs are the **FLICK** trigger (EVENT channel, working) and **biometrics** (HR/RR live + banked gravity/PPG/SpO2/ temp via offload) for the dashboard. The banked **gravity vector** is a genuine orientation signal we can use for coarse wrist-pose checks.

9.10 Gesture control pipeline (gesture → safety → vehicle)

The end-to-end control path, built so the actuator backend is swappable and the safety gate is identical no matter what's on the other end.

```
band flicks ■■                               ■■ MockActuator (today, no car)
               ■■ gesture recognizer ■ SafetyFSM ■ dispatch ■ TeslaFleetActuator (discrete)
keyboard sim ■ (fail-closed, must ARM)         ■■ (future) 2D car / UE5 for DRIVE
```

Gesture vocabulary (small + reliable, given the band's event set):

- **double-flick** = ARM (open the command menu) / CONFIRM the current selection
- **single flick** = cycle to the next command
- **shake** (≥5 impulses) = PANIC: disarm instantly + send LOCK
- **inactivity** (--arm-timeout, default 12 s) = auto-disarm

Safety gate: while IDLE (disarmed) *no* vehicle command can fire — verified in the sim (a flick before arming is ignored). Built on `safety_state_machine_PWT.py` (fail-closed). The menu is **discrete**, **non-motion** commands only.

Why no gesture-driven motion/Summon: Tesla exposes no third-party API for vehicle motion — Smart/Actually/Dumb Summon live only inside Tesla's own app behind its proximity + continuous-press interlocks, with zero programmatic hook. So "flick → car drives to me" has no endpoint to call, regardless of latency tolerance or failsafes, and faking those interlocks is out of scope. The DRIVE demo (continuous hold → motion) is therefore built against a **simulator** (2D first, then UE5), which also proves the continuous-hold loop more cleanly.

Files:

- `vehicle_actuator_PWT.py` — VehicleActuator base, MockActuator, TeslaFleetActuator (REST via the signed-command proxy). Discrete command table; motion/unknown commands are rejected. Self-test passes.
- `gesture_control_PWT.py` — orchestrator. --keyboard runs the full pipeline with no band and no car; band mode recognizes flicks live; --backend tesla [--dry] targets a real car.
- `docs/tesla_fleet_setup_PWT.md` — honest Fleet API setup (dev app, hosted public key, partner + virtual-key pairing, OAuth, tesla-http-proxy signing, env vars).

Status: ■ **WORKING** — **gesture-confirmed UNLOCK executed on the real car (2026-10-05)**. Full chain proven end to end: gesture → fail-closed safety FSM → dispatch → tesla-http-proxy (signs with `private-key.pem`) → Tesla Fleet API → 2018 Model 3. The car negotiated sessions with `DOMAIN_INFOTAINMENT` + `DOMAIN_VEHICLE_SECURITY` and returned **200 OK** on the signed `door_unlock`. Setup that got here: Fleet API app + OAuth (authorization-code), domain + hosted EC public key registered via `partner_accounts`, virtual key paired to the car, signed commands via the local proxy (`TESLA_FLEET_BASE=https://localhost:4443`), **VIN** (not the Fleet API id) as the vehicle tag in the command path.

Gotchas hit + fixed along the way: client secret truncated on hand-copy (use a verified full copy); partner token must drop `openid/offline_access` scopes; GitHub user-site Pages needs `.nojekyll` or the well-known key 404s; `go install` fails on this repo's replace directives (clone + `go build` instead); Git Bash mangles `-subj "/CN=..."` (prefix `MSYS_NO_PATHCONV=1`); the signing proxy requires the

VIN in the path, not the numeric id.

Next: wire real band FLICKs into this (replace keyboard sim), extend the menu (lock/flash/honk/frunk/climate), the DRIVE-by-hold simulator demo, and gravity-pose arming.

9.11 Ergonomics + haptics + one-command launch

Tap, don't flick. Hard wrist flicks hurt, and the band's accelerometer uses firmware wake-on-motion power management we can't override over BLE (keeping realtime HR on does NOT keep motion sampling awake — proven: HR streamed continuously while only one motion event fired, on the tap). The ergonomic fix: **tap the band face** (same 0x0e events, far less strain). First tap firm to wake the accelerometer, then light taps register. Calibration on the user's wrist showed noise under ~1600 and intentional light taps ~2400+, so the default --strength is 2200; --calibrate prints strengths with no commands sent.

Flick/tap recognizer rebuilt with two-level timing: a refractory window (--flick-gap 0.28 s) collapses one tap's ringing into a single tap, then a --multi-window (0.6 s) groups distinct taps → 1 = FLICK, 2 = DOUBLE_FLICK, 3+ = SHAKE. This fixed "a single flick read as a double, a double read as two singles."

Haptic feedback. The WHOOP 4.0 has a buzz motor; drive it via RUN_HAPTICS_PATTERN (opcode 79, body [pattern, loops, 0, 0, 0], pattern 2 = the 4.0 buzz; from whoop-rs haptic.rs). The controller now buzzes on connect / arm / confirm / panic, so the loop is closed: tap to arm → feel it arm → tap to confirm → feel it fire. ble_haptic_PWT.py tests the motor (--sweep compares patterns); --no-haptics disables it.

One-command launch. launch_PWT.ps1 (repo root) refreshes the token, starts the signing proxy if it isn't up, waits for it to listen, sets the env vars, and launches gesture control — then stops the proxy it started on exit. Run: powershell -ExecutionPolicy Bypass -File launch_PWT.ps1, tap to wake, tap-code to act. Full run/tuning/setup instructions live in docs/QUICKSTART_PWT.md.

10. Regenerating the PDF / Word versions

This Markdown file is the master. The PDF and Word copies are generated from it with a small pure-Python script (no LibreOffice needed, so it runs on your Windows machine too). One-time setup:

```
python -m pip install reportlab python-docx
```

Then, from the repo root, regenerate BUILD_LOG_PWT.pdf and BUILD_LOG_PWT.docx any time you edit this file:

```
python Tools/export_docs_PWT.py
```

Re-run it whenever you update this log so the shareable copies stay in sync.

ProjWT build log — private portfolio project. Keep this file updated.